SIMATS ENGINEERING
ASSESSMENT TOOLS

COURSE NAME: **Theory of Computation** **COURSE CODE:** **CSA1304**

COURSE OUTCOME COVERED

CO4: Construct Context-Free Grammars, generate derivations and parse trees to demonstrate the syntactic structure of strings. (BL : 3)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:100**
Duration : 1 Hour

CLASS TEST

Code of Conduct:

I M.VINAY(192311146) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

M.VINAY
Signature of the student:

SIMATS ENGINEERING
ASSESSMENT TOOLS

1. Design a PDA to accept strings with 'n' occurrences of 'a' followed by 'n' occurrences of 'b', where 'n' is a non-negative integer.
i) $L = \{a^n b^n\}$

2. Convert the given context-free grammar (CFG) into a Pushdown Automaton (PDA) that recognizes the same language.

$E \rightarrow E+E \mid E^*E \mid (E) \mid I$
$I \rightarrow Ia \mid Ib \mid I0 \mid I1 \mid a \mid b$

3. Design a TM to accepts the strings with 'n' occurrences of 'a' followed by 'n' occurrences of 'b', where 'n' is a non-negative integer
i) $L = \{0^n1^n \mid n \geq 1\}$

4. Design a Pushdown Automata for the language representing a's followed by b's and the number of b's must be twice than that of a's.

$$L = \{a^n b^{2n} \mid n \geq 1\}$$

5. Construct a Turing Machine to recognize the language. Write the logic used for the design and the formal definition

$$L = \{a^n b^n c^n \mid n \geq 1\}$$

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

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M.vinay

1)

$L = (a^n\ b^n)$

$M = \{Q, \Sigma, \gamma, \delta, q_0, z_0, F\}$

M = Push down automata

Q = set of states

Σ = input alphabet

γ = stack alphabet

δ = input function

q_0 = initial state

F = final state

8 Given

$L = (a^n, b^n)$

$(n \geq 1)$

$\Rightarrow$ aa bb

2) ×

$E \rightarrow E+E \mid E*E \mid (E) \mid I$

$I \rightarrow a \mid b \mid 0 \mid 1 \mid a \mid b$ ×

3)

$L = (0^n, 1^n \mid n \geq 1)$

$M = (Q, \Sigma, \gamma, \delta, q_0, z_0, F\}$

7 Given

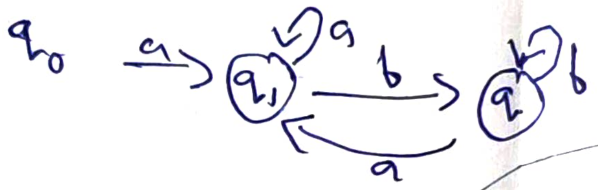
$L = (0^n, 1^n \mid n \geq 1)$

4)

$L = \cancel{\{a^n b^{2m} \mid n \geq 1\}}$

4)

aabbbb   ababbb   a · bab · bb



8

a's followed b's & the no. of b's must be twie then find of a's

$= \{a^n b^{2n} \mid n > 1\}$

3)

$T = \{0, 1, X, Y, B\}$

$S = q_0, q_1, q_2, q_3, q_e, q_r$

Transitions:

$\delta(q_0, 0) = (q_1, X, R)$

$\delta(q_0, X) = (q_0, X, R)$

$\delta(q_0, X) = (q_3, Y, R)$

$\delta(q_0, 1) = (q_r, 1, R)$

8

skip zeros

$\delta(q_1, 0) = (q_1, 0, R)$

$\delta(q_1, Y) = (q_1, Y, R$

$\delta(q_1, 1) = (q_2, Y, 1)$

$\delta(q_3, Y) = (q_0, Y, R)$

$\delta(q_3, B) = (q_1, B, R)$

0011 → XXYY → accept

000111 → XXXYYY → accept

but

00111 → reject

00011 → reject

32/50

5)

1) $L = (a^n b^n c^n \mid n \geq 1)$